A large-scale comparison of cross-situational word learning models

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Author Note

Data, model code, and fitting procedures are available on OSF and in the R package XSLmodels (https://github.com/kachergis/XSLmodels).

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Abstract

One problem language learners face is extracting word meanings from scenes with many possible referents. Despite the ambiguity of individual situations, a large body of empirical work shows that people are able to learn cross-situationally when a word occurs in different situations. Many computational models of cross-situational word learning have been proposed, yet there is little consensus on the main mechanisms supporting learning, in part due to the profusion of disparate studies and models, and lack of systematic model comparisons across a wide range of studies. This study compares 23 models, representing associative, hypothesis-testing, reinforcement-learning, and episodic-memory accounts, on a dataset of 47 experimental conditions with a total of 1,609 adult participants. Using cross-validation, we find that a small group of associative models fit best, that error-driven models do so only with a choice rule sensitive to the scale of the learned associations, and that simulation-based model and parameter recovery can tell most of these mechanisms apart. Finally, we test the models' ability to generalize to seven held-out experiments, including developmental data: fitting well does not predict generalizing well, and every model fit to adults overestimates children's learning.

Keywords: cross-situational word learning; model comparison; language learning; computational modeling

Word count: X

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Introduction

Learning word meanings is a crucial aspect of the process of language acquisition. If a learner's task is to connect the words they hear to the potential referents and eventually to generalizable meanings, the challenge appears formidable (Quine, 1960): words are heard in rich and cluttered contexts. Yet children learn words very quickly. How do they do this?

An array of social (Yurovsky & Frank, 2017), pragmatic (Clark, 1987; Markman, 1992), and linguistic (Gleitman, 1990) cues are often available to reduce a learner's uncertainty. Yet at its heart, these information sources can be seen as reducing the uncertainty inherent in the task of *cross-situational mapping* (Frank et al., 2009): that is, cross-referencing which referents a word has consistently appeared with across multiple uses (Carey & Bartlett, 1978; Gleitman, 1990). If learners can learn cross-situationally, they might be able to pool information about reference (and hence word meaning) over time and learn even absent perfectly disambiguated input in any particular situation. Can children (or even adults) learn in this way?

The general idea of cross-situational word learning has been adopted into a variety of experimental paradigms for testing the word learning ability of infants (e.g., L. B. Smith & Yu, 2008; Vlach & Johnson, 2013), children (e.g., Akhtar & Montague, 1999; Suanda et al., 2014; Vlach & Sandhofer, 2012), and adults (e.g., Yu & Smith, 2007; K. Smith et al., 2011). In such experiments, participants are typically presented with a series of training trials composed of multiple possible referents (e.g., an array of 2-4 novel objects) and one or more spoken nonce words, and (when possible) instructed to learn the meaning(s) of the presented words. These training trials typically present 10–20 distinct word-referent pairs to be learned across 10–50 trials, presenting each word/referent 3–10 times across a total span of < 10 minutes. Learning is then typically assessed by presenting each word and asking learners to make a forced choice of the best referent, either from the entire set or from a small subset of

referents.

People of all ages are able to learn word-object mappings in such paradigms at a level significantly above chance, but their performance is typically far from perfect in all but the most trivial versions of the task (Yurovsky & Frank, 2017). To test the robustness of such learning, many studies have manipulated training and studied resulting performance. How often mappings are presented (Kachergis et al., 2016; Suanda & Namy, 2012), the interval between appearances of a mapping (Vlach & Johnson, 2013; Vlach & Sandhofer, 2012), and the number of mappings presented per trial (Yu, 2008) all have been shown to affect learning performance. In sum, the evidence is strong that people can learn word cross-situationally: we are somehow able to integrate evidence (co-occurrence of words and objects) across trials and use that evidence to learn and remember new word meanings in cross-situational learning experiments. But what are the psychological mechanisms supporting this process? And critically, do these mechanisms support naturalistic word learning?

Computational models provide one strategy for instantiating proposals about mechanism. Such models can then be applied to naturalistic data to make a first assessment of whether cross-situational learning could be a viable strategy for word learning "in the wild". Accordingly, there has been great interest in creating and testing computational models of this task. These models often represent distinct intuitions of how learning operates. At one extreme, *logical* accounts treat the learner as ruling out meanings that are logically inconsistent with the input (e.g., Siskind, 1996). *Hypothesis-testing* accounts propose that learners entertain a single conjectured referent for a word and revise it only when it is disconfirmed on a later encounter, storing little or nothing about the alternatives (Medina et al., 2011; Trueswell et al., 2013). *Associative* accounts instead assume learners accumulate graded word-referent strengths across trials, whether by simple co-occurrence tallying, competitive normalization (Fazly et al., 2010), attention weighted by familiarity and uncertainty (Kachergis et al., 2012), or prediction-error-driven updating in the tradition of

Rescorla and Wagner (1972). Related *reinforcement-learning* accounts recast the problem as learning the value of choosing a referent for a word from a reward signal (here, whether the chosen referent is present), bridging the propose-and-verify logic of hypothesis testing with graded, error-driven value updates. A further question within the associative family is whether the learning rate is fixed or adapts to the learner's current uncertainty, either item-by-item – as in a Kalman-filter formulation of associative learning (Dayan & Kakade, 2001; Gershman, 2015; Kruschke, 2008) – or at the level of the whole system, with learners more alert on trials containing more uncertain items (Fitneva & Christiansen, 2017).

It is unknown which of these models – or even which broad class of models – best fits the breadth of experimental data. While most models of cross-situational learning have typically been tested against at least a few datasets, to our knowledge there has been no systematic comparison of models across a wide variety of experimental conditions. Identifying the model(s) that best predict human performance is thus an important goal for consolidating this broad literature. More broadly, identifying such models could be an important step in assessing the contribution of cross-situational learning to word learning more generally.

A separate concern is that the experiments these models are usually evaluated on are small, tightly controlled, and drawn from a narrow population. A typical adult study presents on the order of a dozen unambiguously referential nonce words, each repeated a handful of times, to English-speaking undergraduates over a few minutes. Naturalistic input looks very different: word-referent co-occurrences in parent-child interaction are sparse, noisy, massively unbalanced in frequency, and only sometimes about anything present in the immediate scene (Frank et al., 2009; Medina et al., 2011). A model that captures the fine structure of laboratory performance need not recover word meanings from corpus data, and vice versa. We therefore also ask how the same models fare when run on transcribed parent-child interaction and scored against a gold-standard lexicon, both at their

laboratory-fitted parameters and when re-fit to the corpora.

Thus, the goal of our current study is to test the ability of a range of cross-situational word learning models in accounting for a wide range of experimental data. Our goal is to scale up both the number of models evaluated and the number of experiment designs (and participants) being used for evaluation. Different models often contain a range of parameters that must be fit to data, an issue that has hindered previous comparison work, either because the parameters have been allowed to vary freely across multiple conditions, or because model performance is compared on different datasets. Thus, we focus on out-of-sample prediction of human data using cross-validated parameter fits. We then examine how well models fit with those parameters generalize to further experiments held out of fitting, including developmental data, and how the models perform on naturalistic parent-child corpora. All of the models, datasets, and fitting procedures are released as an open-source R package, `XSLmodels`, so that the comparison can be extended and reproduced.

Method

Procedure

Our goal is to fit a set of models to a set of datasets; members of each of these sets are listed below. Each model is typically defined as a function that takes input data and a group of parameter values (often between 2–4) and then can be evaluated on test trials (typically multi-alternative forced choice test). Each model then returns predictions on each test trial, which can be scored against the "correct" (intended by the design) answer and averaged to produce an analogue to human performance. Our primary outcome measure is the correlation between each model's performance for each item in a specific experimental condition and item-level human performance on that same experimental condition.

Although input data and test trials are features of specific experimental tests, each model has parameters that must be specified. Typically the settings of these parameters are critical to the predictive performance of the model. Hence, to ensure a fair comparison

between models, these parameters must be fit to data. In all cases, we optimize parameter values to minimize the sum of squared error (SSE) between model and human performance, weighting the contribution of each condition's SSE by the number of participants in that condition. We optimize parameters using differential evolution via the DEoptim R package (Mullen et al., 2011). For the stochastic models, each parameter setting was evaluated using a simulated sample of 200 learners.

We conducted two evaluations of models. First, we optimize each model's parameters on 80% of data and evaluate on the remaining 20% (5-fold cross-validation). Second, in the *group fit* regime, we fit each model to all conditions and datasets and evaluate across all data. Though slightly overfit, this evaluation allows us to make the best test of generalization to other experiments from the literature.

All models, datasets, and fitting procedures are implemented in the open-source R package `XSLmodels`[1], and every fit reported here was produced with version 0.4.0 of that package. Porting the models from the research code behind an earlier conference report of this comparison surfaced several implementation errors, which we corrected, so some results here differ from that report. Most consequentially, the routine shared by the familiarity- and uncertainty-biased family for seeding initial associations to newly familiar words and objects placed those values in the wrong cells whenever some object had not yet been seen; this affected FU and all of its variants. In addition, the guess-and-test model never discarded a hypothesis once formed (its disconfirmation step was unreachable), the probabilistic associative model's test-time accuracy was not normalized over referents, and the Bayesian decay model's choice rule normalized over words rather than over referents. When several words on one trial need new hypotheses at once, the propose-but-verify model now draws each word's proposal independently from the referents present, whereas the earlier

[1] https://github.com/kachergis/XSLmodels; the exact package version used for the fits is recorded with the analysis code.

implementation drew distinct referents for them. Finally, we gave every variant of the FU model the same bounds on the parameters they share with FU (χ, λ, α), so that differences among the variants reflect their mechanisms rather than their search ranges.

Models

We fit twenty-three models together with a zero-parameter baseline co-occurrence model. The original evaluation included the baseline, six associative models (the probabilistic associative model, the familiarity- and uncertainty-biased model, its strength-, uncertainty-, and novelty-biased reductions, a Bayesian decay model, and a Rescorla-Wagner model), two hypothesis-testing models, and two hybrid models that store multiple graded hypotheses for only a subset of the possible associations. We later added eight more: four variants of the familiarity- and uncertainty-biased model (making its learning rate depend on uncertainty at the trial or item level, giving familiarity a free exponent, or adding learning by elimination), a Kalman-filter generalization of the Rescorla-Wagner model, the reinforcement-and-inference model of P. F. C. Tilles and Fontanari (2013), a reinforcement-learning model, and an episodic memory model (MINERVA 2). Finally, we added three error-driven models that follow the original Rescorla-Wagner and Kalman-filter formulations more closely, each with a softmax choice rule, together with a control that applies the same choice rule to co-occurrence counts. All models and their free parameters are briefly described below.

Baseline Co-occurrence Model

The baseline model simply tallies the co-occurrences of any words and objects appearing together on each trial, accumulating the counts in a word $\times$ object matrix M. At test, for word w_1 this model selects the correct referent o_1 according to Luce (1959) choice from the co-occurrences of w_1 with all referents: $P(o_1|w_1) = M_{w_1,o_1} / \sum M_{w_1,\cdot}$. The model's probability of correctly choosing each word per condition is scored against people's average probability of choosing the correct item.

Probabilistic Associative Model (PA)

The probabilistic associative model (Fazly et al., 2010) represents the meaning of each word w as a probability distribution $p(\cdot|w)$ over the objects appearing across the trials, and incrementally learns these distributions trial-by-trial. The association between a given w and o grows more quickly if $p(w|o)$ is already high (a familiarity bias), unless some other word has a strong association with o; thus, associations are competitive. The parameter λ is a small smoothing constant added to both the numerator and denominator of each $p(w|o)$, further adjusted in the denominator by a β parameter representing the upper bound on the number of symbol types. At test, the correct referent for each word w is chosen according to Luce (1959) choice from the probability distribution over the objects that appeared with that word $p(\cdot|w)$.[2]

Familiarity- and Uncertainty-biased Model (FU)

The familiarity- and uncertainty-biased associative model (Kachergis et al., 2012) assumes that learners associate all presented words with all visible objects to some extent, but that they selectively attend more to some of the presented word-object pairings. Specifically, greater attention is directed to pairings that have previously co-occurred (a familiarity bias), but is also directed to stimuli that have no strong associates, tracked via the entropy (H) of their association strengths (e.g., novel stimuli, or stimuli that have diffuse associations with several stimuli). Thus, the model grows its association matrix as it experiences each trial, dynamically apportioning a fixed amount of associative weight (learning rate χ) among the possible word-object associations, with the relative weight of the familiarity bias and the uncertainty bias determined by parameter λ. Association strengths decay at a rate controlled by parameter α. The update rule for adjusting the association $M_{w,o}$ between a given word w and object o on a given trial is:

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[2] The original model implemented a θ-thresholded lexicon that discretized the probability matrix, but we found that including this mechanism resulted in a worse fit to the data.

$$M_{w,o} = \alpha M_{w,o} + \frac{\chi \cdot e^{\lambda \cdot (H(w)+H(o))} \cdot M_{w,o}}{\sum_{w \in W} \sum_{o \in O} e^{\lambda \cdot (H(w)+H(o))} \cdot M_{w,o}} \quad (1)$$

At test, given word w learners use Luce choice, choosing referent o from the m given alternatives in proportion to associative strength $M_{w,o}$.

We also fit a *stochastic sampling* (**FUs**) version of this model (see Kachergis & Yu, 2017), which uses the same update equation, but samples only a single presented object for each word on a trial instead of updating all possible associations. This FU sampling model can still build multiple graded associations for each word (or object), but on any given run will accumulate only a sparse, randomized version of what the full associative model would build.

Strength-, Uncertainty-, & Novelty-biased (Str,Unc,Nov) Models

We also test three variations of the FU model that use only a subset of the mechanisms of the full model in order to understand the contributions of each mechanism. The *strength model* lacks the uncertainty bias term, and thus only implements a bias for familiar (i.e., already-strong) associations, with learning rate (χ) and decay (α) parameters.

The *uncertainty model* lacks the strength bias term, and thus only implements a bias for stimuli with uncertain (high-entropy) associations. The *novelty model* also lacks the strength bias term, and substitutes novelty for the entropy terms (e.g., 1/(frequency(w)+1) instead of $H(w)$). These models have all three parameters of the original FU model, and operate in the same way as that model at test.

Bayesian Decay Model (BD)

This previously-unpublished model updates the $p(w|o)$ and the joint probability $p(w, o)$ from trial to trial according to a likelihood function that reinforces the association between w and o when they co-occur (scaled by parameter δ), and penalizes all associations that are not occurring on the trial (scaled by parameter α). Thus, in contrast to other

incremental associative learning models considered here (e.g., Fazly and the Kachergis class of models), this model globally updates the entire association matrix on each trial – both co-occurring pairs and non-co-occurring pairs, and re-normalizes the tracked probabilities at each time step. At test, this model uses a softmax choice rule with parameter τ to select the best referent for each word from the presented alternatives.

Rescorla-Wagner Model (R-W)

This associative model is inspired by the prediction-error-based learning model of Rescorla and Wagner (1972). The words on each trial serve as cues that predict which objects are present: each object's prediction is the summed association of the trial's words with it. Each present object's prediction error – the difference between the maximum association value λ and its prediction – then changes the association of every word on the trial with that object, at learning rate β, so that the words on a trial compete to predict its objects. Each trial, the association matrix is subject to decay (parameter α). At test, this model uses Luce choice to select the best referent for each word from the presented alternatives. (The version of this model fit in earlier rounds of this comparison misapplied the prediction errors, giving each word the error of a different object, which reduced it to co-occurrence counting; the fits reported here use the corrected model.)

Propose-but-Verify Model (PbV)

In the propose-but-verify hypothesis testing model (Trueswell et al., 2013), a presented referent is selected at random for any presented word that has no remembered referent. The next time that word occurs, the previously-proposed referent is remembered with probability α, a free parameter. If the remembered referent is verified to be present, the future probability of recalling the word is increased by parameter ϵ. If the remembered referent is not present, the old hypothesis is assumed to be forgotten and a new proposal is selected from the available referents. When several words on a trial need new proposals at once, each is drawn independently from the referents present, so two words may propose the same referent.

Guess-and-Test Model (GnT)

The guess-and-test hypothesis testing model is based on the description given by Medina et al. (2011) of a one-shot (i.e. "fast mapping") learning model, which posits that "i) learners hypothesize a single meaning based on their first encounter with a word, ii) learners neither weight nor even store back-up alternative meanings, and iii) on later encounters, learners attempt to retrieve this hypothesis from memory and test it against a new context, updating it only if it is disconfirmed." We give this model two free parameters: a probability of successful encoding (s, hypothesis formation), and a probability f of forgetting a hypothesis at retrieval. At test, for each word the model chooses the currently hypothesized referent.

Pursuit Model (Pur)

The pursuit model (Stevens et al., 2017) is a hybrid model, potentially storing graded associations involving a given word and multiple referents (or vice-versa), while on any given trial greedily pursuing the strongest association for each presented word. If the strongest associated referent for a given word w is present on the trial, the association is strengthened at a rate determined by γ. If the strongest referent for w is not present, the association is weakened (scaled down by $(1-\gamma)$) and the association with a random other available referent is strengthened. Novel words are given an initial association of strength γ with the available referent whose strongest association is the smallest, implementing an uncertainty bias in forming associations for novel words. A given word-object association only enters the lexicon if $p(o|w) > \theta$, a threshold parameter. At test, this model chooses the best referent for each word by Luce choice from the lexicon, which is not necessarily 1-to-1.

The models below were added after the initial comparison.

FU with Trial-Level Attention (FU-A)

This variant of the FU model implements the "system-level" attention account of Fitneva and Christiansen (2017), on which learners are more alert – allocating a higher

learning rate – on trials whose items are more uncertain (for instance early in training, or after many items have been mispaired). On each trial, the associative weight the FU model distributes (learning rate χ) is additionally scaled by the ratio of that trial's mean object entropy to the mean entropy of all objects seen so far. The model has the same three free parameters as the FU model (χ, λ, α) and uses the same Luce choice rule at test.

FU with Item-Level Prediction Error (FU-P)

This variant replaces the FU model's normalization step – which distributes a fixed total amount of associative weight across the word-object pairings on a trial – with an item-level, Rescorla-Wagner-style (Rescorla & Wagner, 1972) prediction error term: the amount learned about a given word-object association is scaled by how far the word's predicted strength (summed over the trial's objects) falls short of a maximum value. Surprising (e.g., initially mispaired) items therefore draw more learning than the normalized model allows. Because there is no normalization step, associations are kept in their valid range by flooring them at zero and capping them at their maximum value. The model has the same three free parameters as the FU model and uses Luce choice at test.

FU with a Familiarity Exponent (FU-G)

In the FU model, the attention a pairing receives on a trial is the product of an uncertainty term, $e^{\lambda(H(w)+H(o))}$, whose influence is set by the free parameter λ, and the pairing's current strength $M_{w,o}$, which enters linearly with no free weight of its own. This variant raises the familiarity term to a free power γ, replacing $M_{w,o}$ with $M_{w,o}^{\gamma}$ in both the numerator and the denominator of Equation 1. Setting $\gamma = 1$ recovers the FU model; $\gamma < 1$ gives diminishing returns to already-strong pairings, dampening the model's "rich get richer" dynamic, while $\gamma > 1$ amplifies it toward winner-take-all. The model has four free parameters (χ, λ, α, γ) and uses Luce choice at test.

FU with Learning by Elimination (FU-E)

This variant adds inference by elimination (cf. Halberda, 2003) to the FU model's allocation of attention. If the other words and objects on a trial are already confidently

paired with each other, the remaining word and object probably go together, so their pairing should draw attention even while it is still weak; in the FU model, by contrast, already-strong pairs on a trial draw attention *away* from such a pairing. Each pair (w', o') on the trial has a mutual confidence $q(w', o') = P(o'|w')\,P(w'|o')$, computed from the row- and column-normalized associations, and a pairing (w, o) receives elimination support $s(w, o) = \sum_{w' \neq w,\, o' \neq o} q(w', o')$ – how much of the rest of the trial is already accounted for. The share of the trial's associative weight given to (w, o) is then $\Big(b(w, o) + \epsilon\, s(w, o)\Big) / \Big(1 + \epsilon \sum s\Big)$, where $b(w, o)$ is the FU model's normalized share, so $\epsilon = 0$ recovers the FU model. The model has four free parameters (χ, λ, α, ϵ) and uses Luce choice at test.

Kalman Filter Model (KF)

The Kalman filter model (Dayan & Kakade, 2001; Gershman, 2015; Kruschke, 2008) generalizes the Rescorla-Wagner model's error-driven update by tracking each word-object association as a Gaussian belief – a mean strength $\mu_{w,o}$ and a variance $\sigma^2_{w,o}$ representing the learner's uncertainty about it. On each trial, an association's update toward its target is scaled by that association's *Kalman gain*, $\sigma^2_{w,o}/(\sigma^2_{w,o} + \sigma^2_{obs})$, which makes the effective learning rate adaptive: large while an association is still uncertain and automatically shrinking as evidence accumulates. Between trials, every association's variance grows by τ^2, so the model can still revise a well-learned belief rather than freezing. The model has three parameters – process noise τ^2, observation noise σ^2_{obs}, and initial uncertainty σ^2_0 – of which only two are identifiable (scaling all three by a constant leaves the learned matrix unchanged). At test, it uses Luce choice over the mean association matrix.

Error-Driven Models with a Softmax Choice Rule (R-W-S, R-W-D, KF-F) and Their Control (Base-S)

Under Luce choice, the R-W and KF models' predictions change little with their learning parameters (Appendix C), so we added three variants that follow the original formulations more closely and score the learned associations with a softmax choice rule,

$P(o \mid w) \propto \exp(\phi\, M_{w,o})$, with inverse temperature ϕ as a free parameter. *R-W-S* is the R-W model above with this choice rule. *R-W-D* also treats the objects absent from a trial as outcomes with target 0, as Rescorla and Wagner (1972) did for absent outcomes, so that a word's associations with objects that fail to appear are unlearned. *KF-F* is a Kalman filter (Dayan & Kakade, 2001) that keeps the full covariance among the words predicting each object, again with absent objects as outcomes; because words that co-occur acquire correlated uncertainty, a word's association with an object can change on trials where that word is not heard, as in backward blocking. In all three, the cue salience, the asymptote λ, and the observation noise are fixed at 1, since they only rescale the other parameters, leaving three free parameters each: β, the decay α, and ϕ for R-W-S and R-W-D; the process noise τ^2, the initial uncertainty σ_0^2, and ϕ for KF-F. *Base-S*, the baseline co-occurrence model with the same softmax rule (one parameter, ϕ), is their control: it shows how much of any improvement comes from the choice rule alone.

Reinforcement-and-Inference Model (TF)

The model of P. F. C. Tilles and Fontanari (2013; see also P. F. Tilles and Fontanari, 2012) represents each word's meaning as a probability distribution over the objects seen so far, updated by two processes. *Reinforcement*, with strength x, transfers confidence for each word heard on a trial from the objects absent from that trial to those present. *Inference*, with strength b, implements a form of mutual exclusivity: a novel word assigns extra probability to the novel objects present on its first trial, and for familiar words *not* heard on a trial, confidence is transferred away from the objects present toward absent ones. Both updates are scaled by a word-specific learning efficiency that starts at α_0 while the word's distribution is maximally uncertain and rises toward 1 as its entropy falls; the remaining share of each update relaxes the distribution toward uniform, acting as noise. The model has three free parameters (x, b, α_0) and uses Luce choice over these distributions at test.

Softmax Reinforcement Learning Model (RL)

The reinforcement learning model recasts word learning as learning, from a reward signal, the value of choosing each referent for a word. Unlike every other model considered here – all of which update every word-object pair that co-occurs on a trial – this model makes a discrete choice: for each word heard, it samples a single candidate referent from a softmax policy over its current values for all known objects (its current best guess at that word's meaning), analogous to the way the propose-but-verify and pursuit models commit to a single hypothesis before checking it. It then observes whether that referent is present in the scene (reward = 1) or not (reward = 0) – the only feedback available in a task with no trial-by-trial correct answer – and updates only the chosen value by a delta rule, $Q_{w,o} \leftarrow Q_{w,o} + \eta\,(\text{reward} - Q_{w,o})$. It has two free parameters: the learning rate η and a softmax inverse temperature β governing how sharply choices favor the highest-valued referent. At test, the model's probability of choosing the correct referent for a word is read directly from the same softmax policy it learned with. This model thus shares the propose-and-verify logic of the hypothesis-testing models but replaces their all-or-none keep/discard rules with graded value learning and explicit, tunable exploration.

MINERVA 2 Episodic Memory Model (M2)

Unlike every other model here, each of which maintains a single word × object association matrix, MINERVA 2 (Hintzman, 1984, 1986) stores each training trial as a separate episodic trace. Words and objects are represented as random feature vectors, and a trial's trace concatenates the sum of its words' vectors with the sum of its objects' vectors; memory is exact, with no decay or forgetting. At test, a word probe activates every trace in proportion to the cube of its similarity to that trace's word half, and the activation-weighted sum of all traces – the "echo" – is compared on its object half with each candidate object. These similarities are exponentiated by a single free parameter X, which sets how sharply choice favors the best-matching object, and normalized as in Luce choice. The dimensionality of the feature vectors (100) is a fixed representational choice rather than a fitted parameter;

because the vectors are random, the model is stochastic. Two further distributed and episodic memory models, TODAM (Murdock, 1982) and REM (Shiffrin & Steyvers, 1997), are implemented in `XSLmodels` but are 20–40 times more expensive to evaluate, which put fitting them at the scale of this comparison out of reach; we leave them to future work.

Data

The modeled data are average accuracies from 924 word-object pairs in 47 experimental conditions, in which a total of 1,609 subjects participated. A table containing details of each condition is available on OSF[3], along with the full trial orderings and aggregate human performance. Each condition consists of an ordered list of 18–108 training trials, each presenting 1–4 words and 2–4 objects, with a total of 12–24 word-object pairs to be learned per condition. At test, participants heard each word and were asked to select the best referent from either all m presented objects (m-alternative forced choice; AFC), or from a subset (e.g., 4AFC). The bulk of these data have been previously published: data from 13 of the conditions were reported in Kachergis and Yu (2013), data from 12 conditions are from Kachergis et al. (2012), and data from another nine conditions are from Kachergis et al. (2016). Data from several conditions were contributed by Chen Yu's lab at Indiana University, and have not been published before. To 42 of the 44 conditions of the original comparison we added five from Kachergis et al. (2009), which manipulated the temporal contiguity of repeated word-object pairs. All 47 conditions tested adults. Experiments that report only condition-level accuracy, or that tested children, are used only to test generalization (see Generalization Experiment Results).

Naturalistic Corpora

To ask whether any of these models could plausibly learn from the input children actually receive, we also ran them on two transcribed corpora of caregiver speech to young children, bundled with XSLmodels. The **Rollins corpus** (Rollins, 2003) is the CHILDES

[3] Information about the dataset: https://osf.io/REDACTED

corpus that Frank et al. (2009)'s Bayesian model was originally fit to: 619 caregiver utterances to 9-month-olds, 416 word types, 22 co-present object types, and a curated gold-standard lexicon of 34 word-object pairs. The **FM corpus** (Frank et al., 2013) is larger and more naturalistic: 4,763 utterances, 1,122 word types, 30 object types, and a gold lexicon of 41 curated pairs (the corpus also ships stricter and more permissive gold-lexicon variants, not used here). Unlike the laboratory conditions above, neither corpus has human referent-selection data to compare against; instead, a model's learned word-object matrix is scored against the gold lexicon directly, as the best-threshold F-score over precision and recall (sweeping the matrix threshold that separates a "learned" pairing from noise). Because a large fraction of caregiver utterances are not about anything in the immediate scene at all, we also ran a Bayesian model built for exactly this setting – Frank et al. (2009)'s joint intentional inference over the whole corpus, optionally extended with a Rational Speech Act pragmatic listener [`fgt2009()`/`fgt2009_rsa()`; not part of the trial-by-trial models above, since it has no learning trajectory to compare against per-condition human accuracy].

Results

Table 1 summarizes each model's fit. Because parameters were fit by minimizing each condition's squared error, weighted by its number of participants, we report that same quantity for the cross-validated (CV) fits – in which each condition is scored once, at parameters fit without it – and for the group fit to all 47 conditions. The best CV fits were from PA (SSE = .483), FU-G (.524), and FUs (.530), whose errors on the five folds' held-out conditions overlap substantially. The order within this top group depends on how conditions are weighted: ranking by squared error summed over all 924 items, ignoring sample size, puts PA first and FUs second. Compared with the zero-parameter baseline co-occurrence model (CV SSE = 1.628, $r = .34$), 19 models fit better by more than 0.01 (PA, FU-G, FUs, KF-F, FU-A, FU-E, FU, R-W-S, TF, Nov, R-W-D, FU-P, Unc, M2, BD, Base-S, Str, RL, and Pur), KF and R-W fit within 0.01 of it, and PbV and GnT fit worse. With Luce choice, neither error-driven model improves on counting. The Rescorla-Wagner model's predictions are

nearly identical to the baseline's (CV SSE = 1.633; item-level $r > .999$ with the baseline's), and in simulation its learning rate could not be recovered nor the model distinguished from the baseline (Appendix C). The Kalman filter (CV SSE = 1.628) is best fit at parameters where learning is so slow relative to the associations' asymptote that each co-occurrence adds a nearly constant increment, so that it, too, reduces to tallying co-occurrences. With a softmax choice rule, however, error-driven learning helps substantially: KF-F, R-W-S, and R-W-D (CV SSE = .571, .608, and .653) all fit far better than their control, Base-S (.889), which shows that the gain is not due to the choice rule alone. Correlations between data and models are presented in Figure 1.

Table 2 shows the correlations between model predictions made using the group-optimized parameters, with the strongest correlation per row in bold.

Generalization Experiment Results

To test whether the fitted models generalize, we ran each model at its group-fit parameters – without refitting – on 7 further experiments, none of which is among the 47 fitting conditions. They are scored at three levels, depending on what the original studies report. *Learning trajectories*: Medina et al. (2011) had adults guess a referent on every trial and reported the proportion of correct guesses in each of five blocks, in four conditions that varied when the informative trials occurred; a model's analogue is its probability of choosing the correct referent among the trial's objects, given what it learned on earlier trials. *Item-level accuracy*: Benitez et al. (2020) compared massed, interleaved, and unstructured naming schedules in children and adults (8 words, 2AFC test; 7 conditions), and an unpublished experiment of ours (Kachergis, Grimmick, & Gureckis) varied how many words were first shown with an object other than their eventual referent (2 conditions). *Condition means*: Koehne et al. (2013) (four orders in which adults met each word's two referents), Suanda et al. (2014) (contextual diversity in 6-year-olds), L. B. Smith and Yu (2008) (12- and 14-month-old infants, 6 words), and Vlach and DeBrock (2017) (spaced and massed

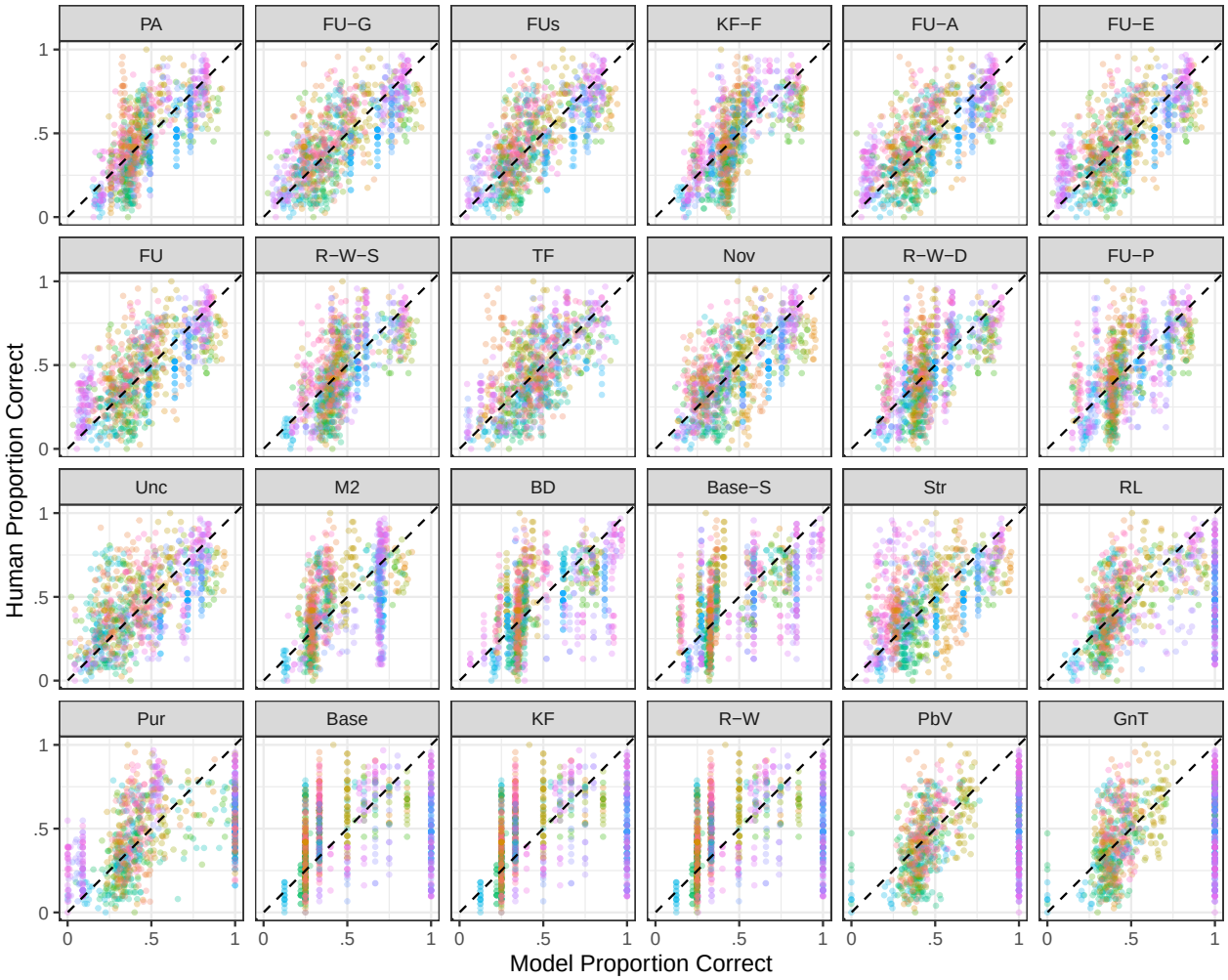


Figure 1

Human vs. model item-level accuracy at each model's group-fit parameters, for all conditions (colored by condition); panels in order of cross-validated fit (Table 1). Note that on some items, PbV, GnT, and Pur all show more extreme responding (1 or 0) than people ever do.

Table 1

Model fits, sorted by cross-validated (CV) error. SSE is the quantity the fits minimize: each condition's squared error summed over its items, averaged over conditions weighted by their number of participants. CV SSE scores every condition once, at parameters fit to the other four folds; the range is over the five folds' held-out sets. Group SSE is for parameters fit to all 47 conditions (for stochastic models, from a fresh simulation of 200 learners at the best parameters). CV r is the item-level correlation of out-of-fold predictions with human accuracy; k is the number of free parameters.

Model	Class	k	CV SSE	CV range (folds)	Group SSE	CV r
PA	associative	2	**0.483**	0.375–0.860	0.473	0.70
FU-G	associative	4	0.524	0.374–0.674	0.510	0.72
FUs	associative (sampling)	3	0.530	0.405–0.741	0.518	0.71
KF-F	associative	3	0.571	0.475–0.691	0.536	0.65
FU-A	associative	3	0.585	0.463–0.717	0.560	0.69
FU-E	associative	4	0.587	0.452–0.727	0.555	0.69
FU	associative	3	0.587	0.452–0.728	0.555	0.69
R-W-S	associative	3	0.608	0.449–0.951	0.566	0.61
TF	associative	3	0.632	0.515–1.120	0.606	0.59
Nov	associative	3	0.638	0.441–0.848	0.620	0.66
R-W-D	associative	3	0.653	0.536–0.954	0.613	0.58
FU-P	associative	3	0.702	0.503–1.025	0.663	0.55
Unc	associative	3	0.730	0.592–0.971	0.678	0.61
M2	episodic memory	1	0.802	0.478–1.183	0.795	0.51
BD	associative	3	0.830	0.641–1.325	0.799	0.54
Base-S	baseline	1	0.889	0.646–1.436	0.876	0.52
Str	associative	2	0.957	0.589–1.223	0.910	0.51
RL	reinforcement learning	2	1.469	1.005–1.879	1.467	0.43
Pur	hybrid	3	1.490	0.972–2.054	1.468	0.52
Base	baseline	0	1.628	0.967–2.042	1.628	0.34
KF	associative	3	1.628	0.967–2.042	1.628	0.34
R-W	associative	3	1.633	0.967–2.042	1.628	0.34
PbV	hypothesis-testing	2	2.065	1.403–2.745	2.049	0.34
GnT	hypothesis-testing	2	2.088	1.413–2.763	2.086	0.34

repetitions in 2- to 5-year-olds). Table 3 gives each model's error on each experiment: RMSE across blocks or items for the first two levels, and the absolute difference from the human

Table 2

Correlations of models' predictions.

	PA	FUs	FU-G	KF-F	FU	FU-E	FU-A	R-W-S	Nov	R-W-D	TF	FU-P	Unc	M2	BD	Str	Base-S	Pur	RL	KF	R-W	PbV	GnT
PA	-	**.89**	.88	.84	.85	.85	.86	**.89**	.88	.83	.82	.84	.79	.77	.78	.70	.78	.69	.68	.60	.60	.56	.56
FUs	.89	-	**.94**	.74	.91	.91	.91	.76	.89	.66	.69	.71	.83	.74	.65	.68	.64	.74	.64	.58	.58	.49	.50
FU-G	.88	.94	-	.73	**.96**	**.96**	**.96**	.76	.87	.66	.68	.69	.86	.71	.64	.69	.62	.72	.61	.54	.54	.48	.48
KF-F	.84	.74	.73	-	.77	.77	.77	.92	.80	**.93**	.75	.85	.54	.52	.70	.79	.71	.46	.36	.23	.22	.26	.26
FU	.85	.91	.96	.77	-	**1.00**	**1.00**	.75	.86	.66	.67	.66	.75	.61	.57	.71	.53	.72	.49	.42	.41	.34	.35
FU-E	.85	.91	.96	.77	**1.00**	-	**1.00**	.75	.86	.66	.67	.66	.75	.61	.57	.71	.53	.72	.49	.42	.41	.34	.35
FU-A	.86	.91	.96	.77	**1.00**	**1.00**	-	.76	.87	.66	.67	.66	.75	.62	.58	.71	.54	.72	.50	.43	.43	.36	.36
R-W-S	.89	.76	.76	.92	.75	.75	.76	-	.82	**.95**	.72	.86	.64	.62	.82	.80	.77	.50	.48	.37	.37	.46	.46
Nov	.88	**.89**	.87	.80	.86	.86	.87	.82	-	.72	.60	.79	.77	.77	.71	.82	.70	.67	.67	.59	.59	.53	.54
R-W-D	.83	.66	.66	.93	.66	.66	.66	**.95**	.72	-	.73	.89	.52	.51	.84	.73	.82	.37	.38	.25	.25	.37	.36
TF	**.82**	.69	.68	.75	.67	.67	.67	.72	.60	.73	-	.69	.59	.57	.56	.56	.64	.52	.47	.37	.37	.32	.31
FU-P	.84	.71	.69	.85	.66	.66	.66	.86	.79	.89	.69	-	.64	.65	.86	.75	**.92**	.38	.56	.45	.45	.46	.45
Unc	.79	.83	**.86**	.54	.75	.75	.75	.64	.77	.52	.59	.64	-	.81	.65	.55	.67	.62	.77	.73	.73	.68	.68
M2	.77	.74	.71	.52	.61	.61	.62	.62	.77	.51	.57	.65	.81	-	.60	.59	.69	.54	**.95**	.92	.92	.84	.85
BD	.78	.65	.64	.70	.57	.57	.58	.82	.71	.84	.56	.86	.65	.60	-	.59	**.93**	.33	.58	.51	.51	.60	.60
Str	.70	.68	.69	.79	.71	.71	.71	.80	**.82**	.73	.56	.75	.55	.59	.59	-	.63	.42	.43	.33	.32	.33	.34
Base-S	.78	.64	.62	.71	.53	.53	.54	.77	.70	.82	.64	.92	.67	.69	**.93**	.63	-	.27	.66	.57	.57	.62	.61
Pur	.69	**.74**	.72	.46	.72	.72	.72	.50	.67	.37	.52	.38	.62	.54	.33	.42	.27	-	.48	.47	.47	.32	.32
RL	.68	.64	.61	.36	.49	.49	.50	.48	.67	.38	.47	.56	.77	.95	.58	.43	.66	.48	-	**.97**	**.97**	.87	.87
KF	.60	.58	.54	.23	.42	.42	.43	.37	.59	.25	.37	.45	.73	.92	.51	.33	.57	.47	.97	-	**1.00**	.88	.88
R-W	.60	.58	.54	.22	.41	.41	.43	.37	.59	.25	.37	.45	.73	.92	.51	.32	.57	.47	.97	**1.00**	-	.88	.88
PbV	.56	.49	.48	.26	.34	.34	.36	.46	.53	.37	.32	.46	.68	.84	.60	.33	.62	.32	.87	.88	.88	-	**.98**
GnT	.56	.50	.48	.26	.35	.35	.36	.46	.54	.36	.31	.45	.68	.85	.60	.34	.61	.32	.87	.88	.88	**.98**	-

mean (averaged over conditions) for the third.

No model generalizes best across the board, and fitting the 47 conditions well does not predict generalizing well. The models' mean ranks across the eight columns of Table 3 are essentially unrelated to their cross-validated fit (Spearman ρ = -.02). Three of the four best mean ranks belong to the models that reduce to counting co-occurrences – KF, the baseline, and R-W – mainly because they are among the closest to the adults of Koehne et al. (2013) and of the initial-accuracy experiment, but they are among the worst on the Medina et al. (2011) trajectories (RMSE = .43, against .20 for the best models, Base-S, FU-P, and BD). The models that fit the 47 conditions best are middling: PA, the best cross-validated model, ranks 12 of 24 on average. No model captures which items were easy in the Benitez et al. (2020) data: no model's item-level correlation with either age group exceeds .41, and most are near zero.

The models consistently overestimate young learners. Fit to adults, every model learns more than the infants of L. B. Smith and Yu (2008) (human accuracy .67; models .83–.98) and the 2- to 5-year-olds of Vlach and DeBrock (2017) (human .56; models

.90–.99), and the median model exceeds the 6-year-olds of Suanda et al. (2014) in all three conditions (by .13–.19). The adult held-out conditions show no comparable bias: on Koehne et al. (2013) the median model is within .10 of the adults' mean in every condition, above it in some and below it in others. Adult-fit parameters thus transfer reasonably well to adults in new designs but not to children, who learn less from the same input than any of the models predicts at adult parameters.

Table 3

Generalization error at the group-fit parameters, on experiments held out of fitting, sorted by mean rank over the eight columns. Medina: RMSE over the four conditions' five blocks. Benitez et al. (adults, children) and initial accuracy: item-level RMSE. Koehne et al. through Vlach & DeBrock: absolute difference between the model's and the learners' mean accuracy, averaged over conditions. Best (lowest) per column in bold.

		Item level							
	Trajectory	Benitez et al.			Condition means				
Model	Medina	Adults	Children	Initial acc.	Koehne	Suanda	Smith & Yu	Vlach & DeBrock	Mean rank
KF	.43	.24	.20	.09	.05	.10	.17	.36	5.6
Base	.43	.24	.20	.09	.05	.10	.17	.36	6.1
FU-P	.20	.24	**.15**	.22	.07	.07	.26	**.34**	6.2
R-W	.43	.24	.20	.09	**.04**	.10	**.17**	.36	6.2
Unc	.23	**.22**	.21	.12	.12	.19	.20	.37	7.8
BD	.21	.23	.17	.26	.05	**.04**	.31	.37	8.1
Base-S	**.20**	.23	.17	.27	.10	.04	.31	.36	8.4
M2	.31	.25	.26	**.08**	.08	.15	.26	.39	9.8
TF	.22	.28	.29	.14	.06	.13	.29	.38	10.8
FU-G	.23	.27	.29	.13	.06	.26	.27	.38	11.5
FUs	.27	.28	.31	.15	.05	.23	.26	.38	12.6
PA	.30	.28	.30	.09	.06	.17	.29	.40	12.9
R-W-D	.24	.24	.22	.21	.16	.07	.31	.38	13.0
RL	.44	.27	.26	.13	.10	.19	.28	.40	14.0
R-W-S	.27	.26	.26	.13	.19	.14	.30	.40	14.0
FU	.24	.28	.31	.13	.12	.26	.29	.37	14.1
FU-A	.23	.29	.31	.15	.09	.26	.29	.37	15.0
FU-E	.24	.28	.31	.13	.12	.26	.29	.37	15.1
KF-F	.29	.26	.26	.17	.22	.13	.31	.39	15.4
Pur	.42	.27	.28	.33	.11	.22	.20	.43	16.2
Str	.35	.29	.30	.14	.32	.18	.29	.39	17.1
Nov	.34	.29	.31	.15	.26	.19	.29	.40	19.0
GnT	.59	.33	.36	.23	.14	.25	.27	.41	20.1
PbV	.58	.34	.36	.21	.13	.22	.30	.41	20.9

Corpus Results

The models above were all fit to maximize agreement with human choices on short, curated laboratory experiments. To ask whether the same mechanisms help recover word meanings from real caregiver speech, we ran every model (plus Frank et al. (2009)'s batch Bayesian model, FGT, and its RSA-pragmatic extension, FGT-RSA) on the Rollins and FM corpora, scoring each model's learned matrix against the corpus's gold lexicon by best-threshold F. We report two regimes: *default* parameters (each model's XSLmodels registry starting values, informed by the lab fits but not tuned to either corpus) and *corpus-fitted* parameters (re-optimized on each corpus to maximize gold-lexicon F). Table 4 gives F for both regimes on both corpora, ordered by mean fitted F.

Fitting to the corpus helps every model, often substantially – unsurprisingly, since registry defaults were tuned on short controlled experiments with none of a corpus's sparsity or non-referential noise. The gain is largest for models whose defaults transfer worst: guess-and-test gains roughly .05-.06 F from fitting, and the item-level-prediction-error BAM variant (FU-P) gains .15-.32, while PA's defaults were already close to optimal for a corpus and barely move. Two findings stand out for the comparison between laboratory and corpus performance (Spearman's $\rho = 0.59$ between each model's lab rank and its corpus rank, computed over the 13 models tested in both settings): first, the Bayesian decay model – middling in the lab comparison – is the strongest model on *both* corpora, and FGT, the model purpose-built for exactly this inference problem, is close behind (its RSA extension makes almost no difference: the learned lexicon is too sparse for a heard word to ever name two co-present objects, so the pragmatic listener collapses to the literal one). Second, the lab-best models diverge in how well they transfer: PA holds up as one of the strongest corpus models, while FUs – the single best-fitting lab model – falls to the middle of the corpus ranking, and the plain FU model it is built on fares only slightly better. The hybrid Pursuit model, already the weakest lab fit, is also the weakest here, collapsing to a near-flat,

uninformative matrix on the larger FM corpus.

Table 4

Best-threshold F recovering the gold lexicon on two naturalistic corpora, at registry-default and corpus-fitted parameters. FGT and FGT-RSA have no lab counterpart. Best per column in bold.

	Rollins		FM		
Model	default	fitted	default	fitted	Mean fitted F
BD	0.45	0.46	**0.45**	**0.56**	0.51
FGT	0.38	0.45	0.42	0.53	0.49
FGT-RSA	0.40	0.45	0.42	0.52	0.48
RL	0.41	0.45	0.26	0.46	0.46
FU-A	0.37	0.44	0.35	0.44	0.44
PA	**0.45**	**0.47**	0.41	0.41	0.44
FU	0.37	0.42	0.35	0.45	0.43
GnT	0.41	0.44	0.30	0.36	0.40
FU-P	0.21	0.36	0.09	0.41	0.39
FUs	0.38	0.38	0.35	0.35	0.36
PbV	0.32	0.36	0.24	0.33	0.35
R-W	0.31	0.35	0.29	0.32	0.34
KF	0.30	0.38	0.21	0.27	0.33
Base	0.38	0.38	0.27	0.27	0.32
Pur	0.35	0.35	0.05	0.05	0.20

Discussion

We set out to conduct a systematic comparison of models of cross-situational word learning across a wide variety of experimental conditions. Using a large dataset, we found best-fitting parameters for 23 models (plus the parameter-free baseline), both by simultaneously fitting all of the data and by cross-validation. The results of both fitting regimes were consistent: as shown in Table 1, the associative models generally achieved better fits (lower SSE and higher correlations) than the hypothesis-testing models (GnT; Medina et al. (2011) and PbV; Trueswell et al. (2013)), whose predictions were almost interchangeable (Table 2: $r = .98$). However, it was also striking how similar many models' predictions were, even for models representing disparate learning theories.

Under Luce choice, the Rescorla-Wagner model's predictions were essentially identical to those of the Kalman filter ($r > .99$) and of the baseline, all three reducing to co-occurrence counting. This is not a coincidence of the fitted parameters: with Luce choice, the R-W model's learning rate barely changes its predictions on these designs, so the two models cannot be told apart even on data simulated from one of them (Appendix C). Given a choice rule sensitive to the scale of the learned associations, however, the error-driven models fit far better than the same choice rule applied to co-occurrence counts, so prediction-error learning may deserve more attention than its Luce-choice versions suggest. Some models' particular patterns of responses suggest theoretical shortcomings. For example, the hybrid Pursuit model (Stevens et al., 2017) fared poorly, in part because it often outperformed humans ($M_{\text{Pur}} = .51$, $M_{\text{hum}} = .46$). Indeed, as shown in Figure 1, Pur, PbV, and GnT all showed a pattern of inhuman responses: for a subset of items, the models either always got them correct or always got them incorrect, while people were more variable. And yet Pur's predictions were most highly correlated with those of FUs ($r = .74$), one of the best-fitting models. In summary, although the hypothesis-testing models evaluated here did not fare well, models from different theories can often mimic each other – at least in some experimental conditions – so it is necessary both to test other instances of general theories (including hypothesis-testing models with, say, a decaying memory to reduce ceiling performance) and to use models to generate predictions during experiment design, in order to choose more informative experiments.

Among the associative models, PA, FU-G, and FUs fit best. Their predictions were more correlated with each other (PA vs. FU-G: .88; PA vs. FUs: .89; FU-G vs. FUs: .94) than with the human data ($r = .70$–$.72$; Table 1). But PA's predictions were as close to those of R-W-S ($r = .89$), a model with a quite different mechanism, as to those of the FU family. Examining the experimental conditions that were easiest for the models to fit can indicate what types of designs are less useful for testing theories of learning. For example, the condition best accounted for by the most models (10 of 23) had varied frequency (pairs

appeared 3, 6, or 9 times) with low contextual diversity (i.e., the words of each frequency co-occurred only with each other): it turns out that showing a performance advantage for more frequent items is straightforward for many theories. The models' worst-fit conditions, by contrast, were spread across designs: the most common, 2x4 6/12 4AFC, was the worst for only 4 models. Future work should aim to further explore when these models make overlapping predictions, but another approach is to find conditions where models make disparate predictions. FU-G and PA make their most divergent predictions in the conditions 2x4 6/12 4AFC, 1x3, and filt3E_3L, which hints at a difference in mechanism between these models. A fruitful next step would be to use optimal experiment design to generate a definitive experiment to differentiate these models.

Although we have aimed for broad coverage of models and experimental conditions, this study is far from complete in both respects. First, there are many other models that we have not yet implemented (e.g., McMurray et al., 2012). Moreover, this dataset represents performance from a fairly homogeneous sample: English-speaking US college students. Future work should aim to include not only more models, but a more broadly diverse sample, both in terms of language background and age. The small number of generalization experiments run here suggest that parameters fitted to adult experimental conditions do not generalize well to developmental experiments: future research should gather a larger set of developmental data and re-fit these models with cross-validation. We invite other researchers to contribute datasets and model implementations for future, larger-scale comparisons: data, model code, and fitting procedures will be open-sourced and available on OSF, as well as through an R package `XSLmodels` which we invite other to contribute to.

In summary, we have shown not only how a large model comparison can help rank models from most to least able to account for human data, but also how such comparisons can identify models that mimic each other, as well as identifying experimental designs that might aid in distinguishing models. Future work should extend this comparison to

cross-situational word learning models and extant datasets beyond those tested here, and design novel experiments designed to distinguish these models. Finally, assuming that cross-situational learning is an important tool in the language learner's toolbelt, it is important to consider how these models can be extended to incorporate other information sources at the learner's disposal, such as social cues, pragmatics, and syntactic constraints on meaning (Hirsh-Pasek et al., 2000; Markman, 1990; Yurovsky & Frank, 2017). While many of these models have mechanisms for preferentially selecting and storing particular associations, most have yet to formally incorporate relevant cues from social partners–linguistic and nonlinguistic (though see Yurovsky & Frank, 2017). More broadly, we believe that structured model comparison is a critical step towards developing more complete models and, eventually, quantitative identification of the fundamental mechanisms of word learning.

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Appendix A

Summary of Experimental Conditions

Table A1 describes the 47 experimental conditions included in the model comparison, including the number of trials, the number of words presented per trial (Words/Trial), the number of referents presented per trial (Objects/Trial), the number of to-be-learned word-referent pairs (Items), people's overall mean accuracy ($p(o|w)$ across all intended $w - o$ mappings) in each condition (Accuracy), the standard deviation of performance (SD), and the number of participants per condition (N).

Table A1

Summary of modeled datasets.

Condition Name	Trials	Words/Trial	Objects/Trial	Items	Accuracy	SD	N
3x4	36	3	4	18	0.19	0.12	25
3x4 1/.5	36	3	4	18	0.22	0.11	25
3x4 1/.66	36	3	4	18	0.21	0.09	25
3x4 +6o	36	3	4	18	0.27	0.12	20
2x4	54	2	4	18	0.30	0.15	33
3x3 +1w/o	54	3	3	18	0.17	0.08	39
4x4 +2w/o	54	4	4	18	0.10	0.05	39
1x3 6/18 4AFC	108	1	3	18	0.67	0.10	43
2x3 6/9 4AFC	54	2	3	18	0.69	0.11	38
2x4 6/12 4AFC	54	2	4	18	0.62	0.12	31
3x4 6/8 4AFC	36	3	4	18	0.69	0.09	36
1x3	108	1	3	18	0.74	0.11	23
2x3	54	2	3	18	0.58	0.07	23
2x4 6/12	54	2	4	18	0.33	0.18	14
3x4 6/8	36	3	4	18	0.42	0.14	13
2x3 6/9	54	2	3	18	0.55	0.11	32
3x4 9/12	54	3	4	18	0.69	0.08	33
1x4	108	1	4	18	0.19	0.10	40
1x3	108	1	3	18	0.39	0.12	40
4x4	27	4	4	18	0.31	0.07	77
3x3	36	3	3	18	0.43	0.08	36
2x2	54	2	2	18	0.79	0.11	19
2x2 5x 24AFC	60	2	2	24	0.51	0.12	46
filt0E_3L	18	2	2	24	0.37	0.11	31
filt3E_3L	27	2	2	24	0.45	0.29	30
filt6E_3L	36	2	2	24	0.42	0.30	30
filt9E_3L	45	2	2	24	0.41	0.30	31
filt0E_6L	36	2	2	24	0.49	0.10	23
filt3E_6L	45	2	2	24	0.48	0.22	23
filt6E_6L	54	2	2	24	0.55	0.27	23
filt9E_6L	63	2	2	24	0.51	0.24	23
filt0E_9L	54	2	2	24	0.59	0.11	31
filt3E_9L	63	2	2	24	0.58	0.27	31
filt6E_9L	72	2	2	24	0.59	0.26	31
filt9E_9L	81	2	2	24	0.60	0.27	31
2_x8_39_4x4	27	4	4	18	0.41	0.16	30
3_x8_369_4x4	27	4	4	18	0.39	0.13	30
freq369-3x3loCD	36	3	3	18	0.49	0.23	34
freq369-3x3hiCD	36	3	3	18	0.56	0.12	104
freq369_36mx	36	3	3	18	0.45	0.16	66
freq369_39mx	36	3	3	18	0.62	0.14	40
cont_div6-12	36	3	3	18	0.43	0.11	62
temporal+spatial contiguity	27	4	4	18	0.49	0.08	30
temporal contiguity only	27	4	4	18	0.40	0.10	32
1 overlap per 3 trials, contr	27	4	4	18	0.32	0.15	31
1 overlap per 2 trials, contr	27	4	4	18	0.28	0.14	31
2 overlap per 2 trials, contr	27	4	4	18	0.41	0.11	31

Appendix B

Model Misfit by Experimental Condition

The heatmap below shows cross-validated model fit (SSE; sum of squared error) for each experimental condition, across all 23 fitted models and the 46 conditions with cross-validated fits. Note that each model has difficulty fitting at least one or more experimental conditions–and these difficult conditions vary somewhat by model. For example, the strength-biased model has particular difficulty with the `filt` conditions[4] (Kachergis et al., 2012), which present a group of word-referent pairs early in training which then systematically co-occur with particular novel late-stage word-referent pairs, testing how strictly learners will maintain a mutual exclusivity (ME) constraint. The Propose-but-Verify model also shows greater misfit in most of these conditions (except for the `filtXE_3L` conditions, which have only 3 repetitions of the late-stage pairings, and thus do not overwhelm learners' ME bias).

On the other hand, many experimental conditions are nearly equally well fit by all models, especially those that have a fixed number of repetitions per word-referent pair (e.g., 3x3 and 4x4, although 2x2 presents difficulties for some models).

Correlation of Model and Human Item-Level Performance per Condition

Each cell displays the correlation coefficient of model vs. human item-level performance in a given experimental condition.

Which Experimental Designs Are Easy or Hard Overall?

The heatmaps above group misfit by model; here we instead average over models, to ask which experimental designs are uniformly easy or hard to predict for *any* model – a useful signal for choosing which conditions actually discriminate between theories, versus which conditions every model gets right (or wrong) regardless of mechanism.

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[4] Except for the `filt0E_` conditions, which consist only of the late-stage pairs, with no early stage.

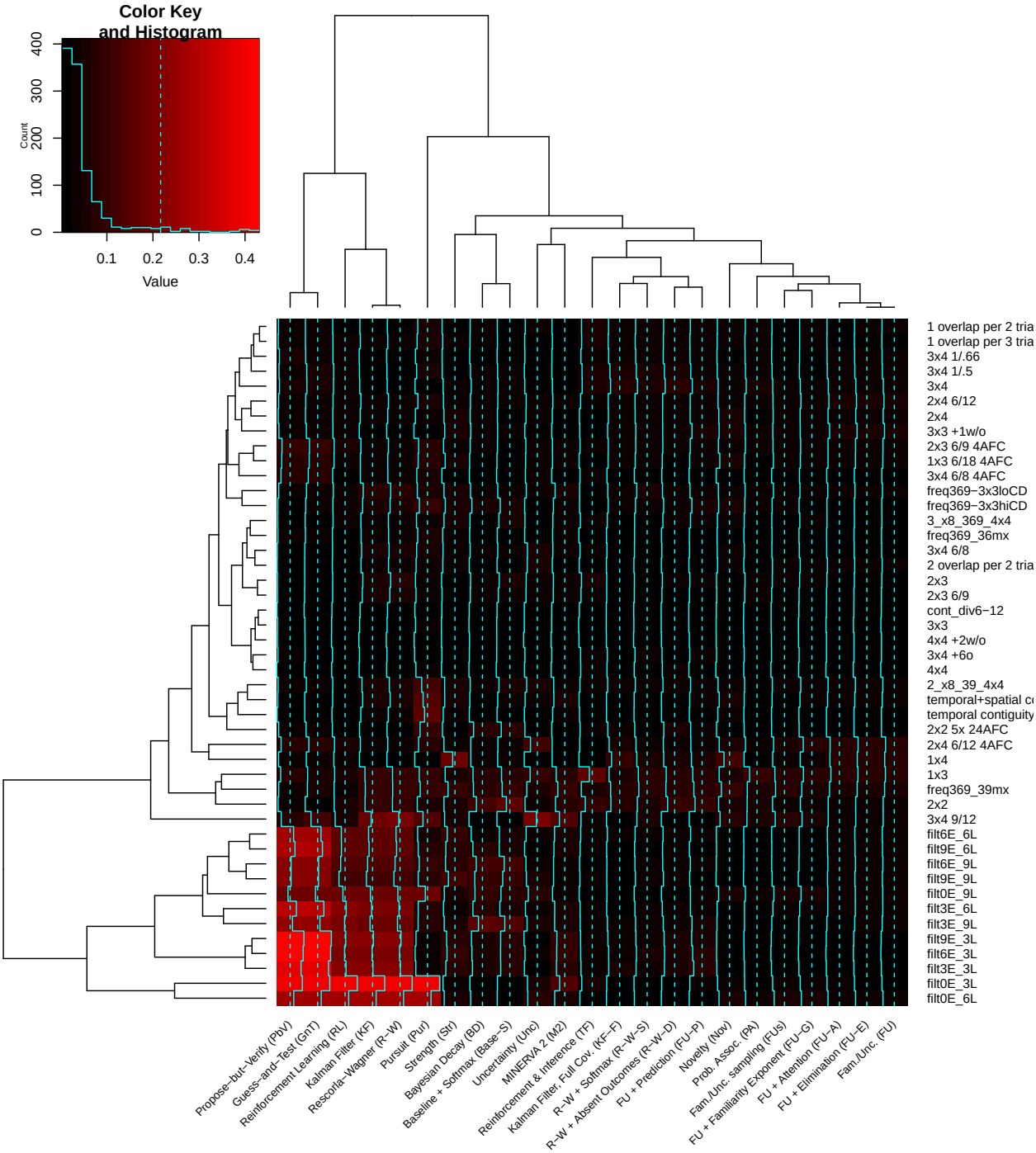


Figure B1

Cross-validated misfit (mean squared error per item) of each model on each experimental condition, clustered by condition (rows) and model (columns); brighter red is worse.

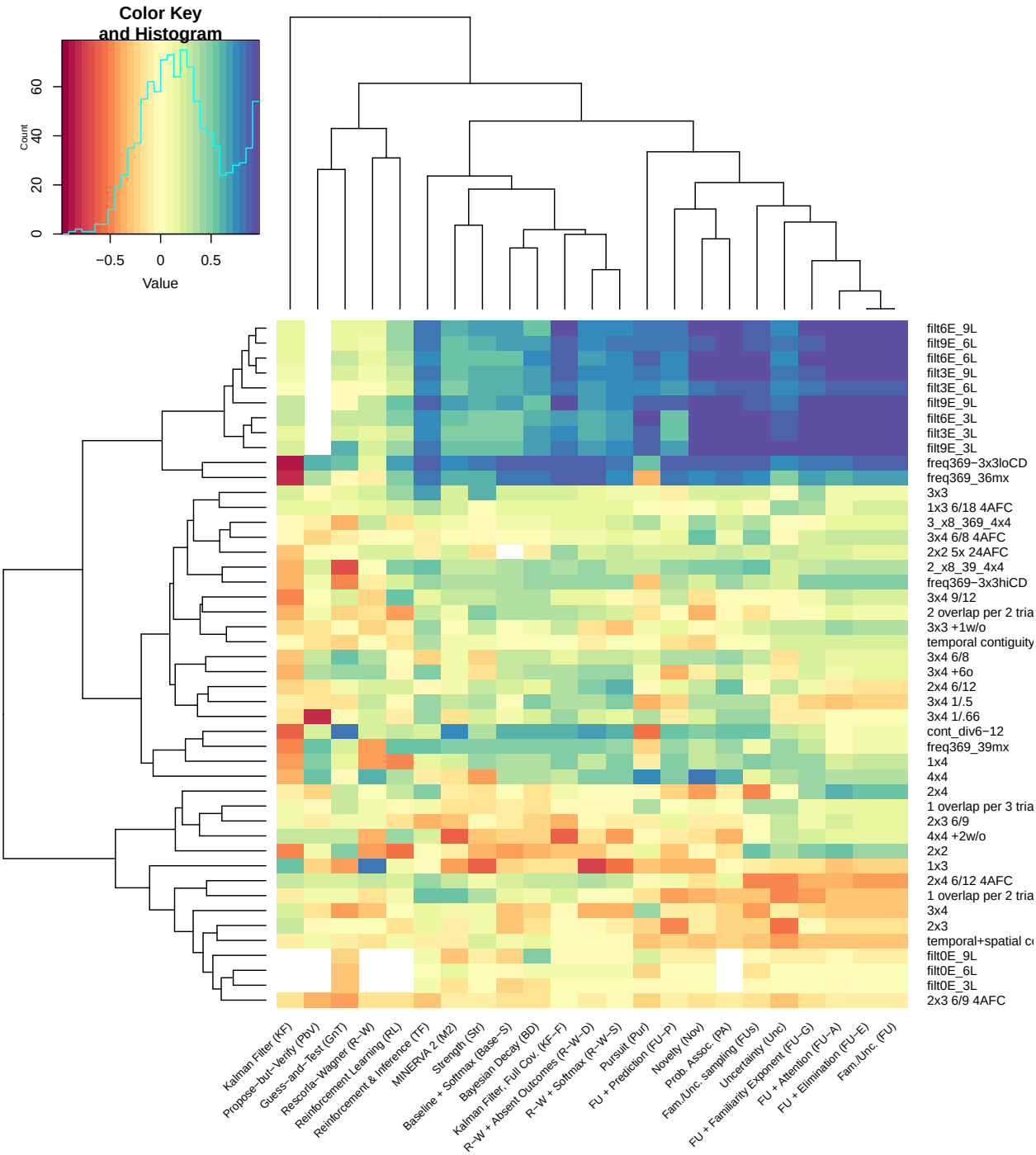


Figure B2

Item-level correlation between each model's cross-validated predictions and human accuracy, per experimental condition, clustered by condition (rows) and model (columns).

Table B1 shows the five conditions with the lowest mean SSE across all 23 models (easiest, fit well by essentially every model) and the five with the highest (hardest, fit poorly by essentially every model). The `SD SSE` column shows how much models disagree on a condition: a low SD alongside a high mean SSE means every model struggles with that design about equally – a design that is hard for principled reasons rather than one that merely discriminates between theories.

Model Disagreement: A Discriminability Index Independent of Human Data

Mean SSE (above) indexes how *hard* a condition is, but hardness and discriminative *value* are different things: a condition every model gets equally wrong is hard but uninformative for choosing between theories, while a condition that splits the models – whoever turns out to be right – is exactly the kind of experiment worth running again with more power, or in a new population. We index this directly, independent of human data, as the standard deviation of each model's predicted item-level accuracy (`ModelPerf`, at each model's group-fit parameters) across models, averaged over the items in a condition. Because several of our associative models are one-parameter-removed cousins of each other (FU/FUs/FU-A/FU-P/Str/Unc/Nov all share the same core mechanism), their mutual spread can pad this index without reflecting genuine disagreement between *theories*. We therefore report it two ways: across all 24 fitted models, and across 12 representatives, one per distinct mechanism (Fam./Unc. (FU), Prob. Assoc. (PA), Bayesian Decay (BD), Rescorla-Wagner (R-W), Propose-but-Verify (PbV), Guess-and-Test (GnT), Pursuit (Pur), Reinforcement Learning (RL), Kalman Filter (KF), Reinforcement & Inference (TF), MINERVA 2 (M2), Baseline) – the two versions correlate at 0.99, but reorder some conditions meaningfully (see below).

Table B2 shows the most- and least-discriminative conditions by this measure. The `filt9E_*`/`filt6E_*` conditions – which present a group of pairs early, then systematically pair a subset of the late-stage objects with the early-stage words to test how strictly a

mutual-exclusivity bias is maintained – are the most model-discriminating conditions in the whole set, on both versions of the index; the temporal- and spatial-contiguity conditions also rise near the top once the FU family's internal spread is removed, indicating a genuine cross-theory split there rather than one associative model disagreeing with its own close cousins. At the other end, `filt0E_6L`/`filt0E_9L` (the late-stage-only `filt` conditions, with no early pairs to violate) and several of the fixed-repetition `3x4`/`4x4` conditions are answered almost identically by every model regardless of mechanism – useful as sanity checks, but uninformative for adjudicating between theories. This index correlates with the SSE-based "hardest" ranking above at only $r \approx 0.75$, confirming that "hard for everyone" and "discriminates between theories" are related but distinct properties of a design.

Each Model's Best- and Worst-Fitting Experiment

Table B3 summarizes for each model, what experimental condition is best fit by that model (in terms of correlation, `best` and `best_cond`), and what condition has the worst fit for that model (`worst` and `worst_cond`).

Across models, freq369-3x3loCD is most often the best-fit condition (best for 10 of 23 models), and 2x4 6/12 4AFC is most often the worst-fit condition (worst for 4 models).

Best- and Worst-Fitting Models per Experiment

Table B4 takes the complementary view: for each experimental condition, which model fits it best and which fits it worst (by item-level correlation), and then, aggregating over conditions, how many conditions each model "wins" (is the single best-fitting model for) and "loses" (is the single worst-fitting model for). A model that wins many, varied conditions is a safe default; a model that wins a few conditions no other model wins is doing something distinctive worth understanding.

Table B1

The five easiest and five hardest experimental conditions, by mean SSE across all fitted models.

Rank	Condition	Mean r	Mean SSE	SD SSE
Easiest	4x4 +2w/o	-0.042	0.011	0.005
Easiest	3x3	0.219	0.011	0.006
Easiest	4x4	0.314	0.012	0.009
Easiest	cont_div6-12	0.338	0.013	0.005
Easiest	3x4 +6o	0.150	0.019	0.006
Hardest	filt0E_3L	-0.043	0.132	0.169
Hardest	filt0E_6L	0.008	0.098	0.107
Hardest	filt9E_3L	0.725	0.096	0.127
Hardest	filt3E_3L	0.671	0.095	0.114
Hardest	filt6E_3L	0.702	0.093	0.121

Table B2

Most- and least-discriminative experimental conditions: standard deviation of model-predicted item accuracy across models, at each model's group-fit parameters. 'All models' is all 24 models including the baseline; 'mechanism reps' is one representative per distinct theoretical mechanism (12).

Rank	Condition	SD (all models)	SD (mechanism reps)
Most discriminative	filt9E_3L	0.345	0.353
Most discriminative	filt6E_3L	0.326	0.337
Most discriminative	filt9E_6L	0.316	0.324
Most discriminative	filt3E_3L	0.304	0.318
Most discriminative	filt6E_6L	0.300	0.304
Most discriminative	filt9E_9L	0.300	0.304
Least discriminative	3x4 1/.66	0.068	0.072
Least discriminative	3x4	0.067	0.070
Least discriminative	cont_div6-12	0.066	0.069
Least discriminative	4x4 +2w/o	0.056	0.067

Table B3

Each model's best- and worst-fitting experiment.

Model	best	worst	best_cond	worst_cond
Fam./Unc. sampling (FUs)	0.98	-0.48	filt6E_3L	2x4 6/12 4AFC
FU + Familiarity Exponent (FU-G)	0.97	-0.42	filt9E_9L	1 overlap per 2 trials, contr
FU + Attention (FU-A)	0.97	-0.38	filt6E_3L	2x4 6/12 4AFC
Prob. Assoc. (PA)	0.97	-0.37	filt9E_9L	4x4 +2w/o
Fam./Unc. (FU)	0.97	-0.40	filt9E_9L	2x4 6/12 4AFC
FU + Elimination (FU-E)	0.97	-0.40	filt9E_9L	2x4 6/12 4AFC
Novelty (Nov)	0.96	-0.41	filt9E_9L	2x4
Uncertainty (Unc)	0.95	-0.53	filt9E_3L	2x3
Pursuit (Pur)	0.95	-0.53	filt6E_3L	cont_div6-12
Kalman Filter, Full Cov. (KF-F)	0.93	-0.63	filt6E_9L	4x4 +2w/o
Bayesian Decay (BD)	0.90	-0.36	freq369-3x3loCD	2x2
Baseline + Softmax (Base-S)	0.90	-0.44	freq369-3x3loCD	2x2
R-W + Absent Outcomes (R-W-D)	0.89	-0.74	freq369-3x3loCD	1x3
FU + Prediction (FU-P)	0.87	-0.48	freq369-3x3loCD	2x3
Reinforcement & Inference (TF)	0.85	-0.39	freq369-3x3loCD	2x3 6/9
Strength (Str)	0.84	-0.59	freq369-3x3loCD	1x3
Rescorla-Wagner (R-W)	0.84	-0.41	1x3	1x4
R-W + Softmax (R-W-S)	0.82	-0.57	freq369-3x3loCD	1x3
Guess-and-Test (GnT)	0.79	-0.69	cont_div6-12	2_x8_39_4x4
MINERVA 2 (M2)	0.77	-0.62	freq369-3x3loCD	4x4 +2w/o
Reinforcement Learning (RL)	0.71	-0.54	freq369-3x3loCD	2x2
Propose-but-Verify (PbV)	0.60	-0.83	freq369-3x3loCD	3x4 1/.66
Kalman Filter (KF)	0.58	-0.90	1x3	freq369-3x3loCD

Table B4

Number of experimental conditions that each model fits best, and worst (by item-level correlation).

Model	best_fits	worst_fits
Reinforcement & Inference (TF)	7	1
FU + Attention (FU-A)	5	0
Kalman Filter, Full Cov. (KF-F)	4	1
Bayesian Decay (BD)	3	0
FU + Elimination (FU-E)	3	0
Fam./Unc. sampling (FUs)	3	2
Novelty (Nov)	3	0
Reinforcement Learning (RL)	3	3
FU + Familiarity Exponent (FU-G)	2	0
Guess-and-Test (GnT)	2	10
Pursuit (Pur)	2	1
Strength (Str)	2	1
Uncertainty (Unc)	2	4
FU + Prediction (FU-P)	1	2
Kalman Filter (KF)	1	10
MINERVA 2 (M2)	1	2
R-W + Softmax (R-W-S)	1	1
Rescorla-Wagner (R-W)	1	5
Propose-but-Verify (PbV)	0	2
R-W + Absent Outcomes (R-W-D)	0	1

Appendix C

Model and Parameter Recovery

Two questions bear on how far the comparison can be trusted. First, if one of these models had generated the data, would fitting all of them have identified it, or do some mechanisms mimic each other so closely on these 47 designs that the comparison could not tell them apart? Second, are a model's fitted parameters interpretable, or could quite different parameter values produce the same predictions? We answered both with simulation, using a reduced version of the fitting procedure in the main text.

Synthetic data. A generating model was run on all 47 conditions (with 500 simulated learners for stochastic models, to approximate its true predictions closely); each item's synthetic accuracy was then drawn as a binomial proportion with that item's real number of participants. Synthetic data are therefore exactly as noisy as the human data under the binomial noise model behind the noise ceiling, though they lack participant-level individual differences.

Fitting. Synthetic data were fit with the same objective (participant-weighted SSE) and parameter bounds as the main analysis, but a reduced differential-evolution budget, so that several hundred fits were affordable: a population of 10 per free parameter (at least 20), and 60 generations for deterministic models or 25 for stochastic ones, whose fits used 50 simulated learners per evaluation. We used one representative of each mechanism: PA, FU, FUs, BD, TF, M2, RL, and PbV, plus the baseline in model recovery.

Parameter recovery. For each of the 8 models, 20 parameter vectors were drawn uniformly within the fitting bounds (log-uniformly for scale parameters whose bounds span two or more orders of magnitude), discarding draws whose mean predicted accuracy fell below .1 or above .95, since such designs carry little information. Each synthetic dataset was then fit by the model that generated it.

Model recovery. Each of the 9 models generated 5 synthetic datasets at its group-fit parameters from the main analysis, and every dataset was fit by all 9 models. The winner was the model with the lowest objective; stochastic models were re-evaluated with 200 simulated learners at their best parameters, since the minimum the optimizer reports for a noisy objective is biased low.

Results

Parameters are mostly recoverable. Figure C1 plots recovered against generating values, and Table C1 summarizes them. 18 of the 19 parameters were recovered with $r \geq .70$ (on a log scale for the log-scaled ones), and 9 with $r \geq .95$. No recovered value fell on a fitting bound. There are three caveats.

- **The propose-but-verify model's recall probability α is not identifiable** ($r = .33$): recovered values cluster near .5 whatever the generating value, plausibly because α and the recall increment ϵ both control how often a proposed referent survives to be verified, so different combinations predict similar accuracy. PbV's fitted α should not be interpreted on its own.
- **PA's smoothing constant λ is identifiable only when it is not negligible.** Above about 10^{-4} it is recovered closely (median error 0.03 $\log_{10}$ units); below that, any small value predicts the same accuracies, so the recovered value is arbitrary.
- **Several parameters are recovered only loosely:** BD's τ (.74), FUs' λ (.78), RL's β (.80), BD's δ (.80), PbV's ϵ (.82), and BD's α (.83). Their occasional large misses (Figure C1) suggest partial trade-offs with the same model's other parameters, so these values should be interpreted with caution.

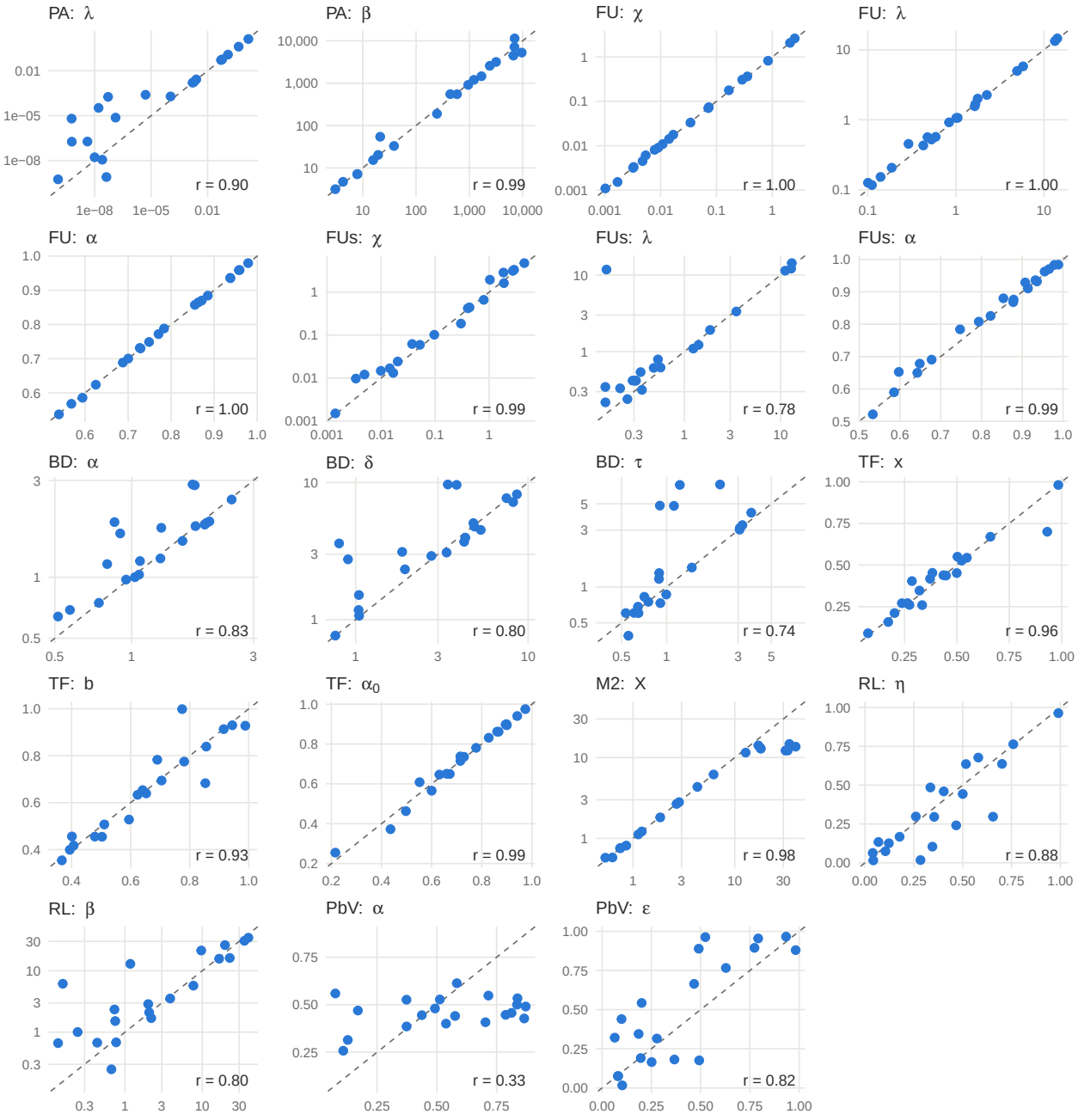
Models are perfectly recoverable at their fitted parameters. In every one of the 45 synthetic datasets, the generating model fit best (45 of 45; Figure C2). For every generator except the baseline, the closest rival's objective exceeded the generator's by at least .07 (Figure C3). The baseline is the hardest case: its data were best fit by the baseline every

time, but by margins of only .01–.02 over the reinforcement-learning model. The closest rivals are mostly the expected ones: FU and FUs for each other; PA for BD, TF and M2; M2 for RL; and RL for PbV, the two models that commit to one referent per word on each trial.

What this does and does not show. Perfect recovery means that the 47 designs, at their real sample sizes, are informative enough to distinguish these mechanisms *when the data sit where each model's best fit to the human data puts them.* It does not show that every region of each model's parameter space is distinguishable, and the reduced fitting budget and in-sample selection make it, if anything, a lenient test. Nor does it make the human-data ranking any less dependent on the models' specific assumptions: a model can be recoverable and still be wrong.

Error-driven models

The Rescorla-Wagner model was refit after a bug was corrected in its update (in the version first fit, each word received the prediction error of a different object, which reduced the model to co-occurrence counting). We also added three error-driven models that follow the original formulations more closely, all with a softmax choice rule at test ($P(o|w) \propto \exp(\phi\, M_{w,o})$, inverse temperature ϕ) in place of Luce choice: R-W-S, the Rescorla-Wagner model with a softmax choice rule; R-W-D, which also treats the objects absent from a trial as outcomes with target 0, as in Rescorla and Wagner (1972), so that associations with absent objects are unlearned; and KF-F, a Kalman filter with full covariance among the words predicting each object and absent objects as outcomes (Dayan & Kakade, 2001), in which a word's association with an object can change on trials where that word is not heard (e.g., backward blocking). In all three, the salience and the asymptote are fixed at 1 and the observation noise at 1, since they only rescale the other parameters; the free parameters are the learning rate β (or, for KF-F, the process noise τ^2 and initial uncertainty σ_0^2), the decay α (R-W-S and R-W-D), and ϕ. Base-S, the co-occurrence baseline with the same softmax rule, is the control: it shows how much of the



x: generating (true) value; y: recovered value; dashed: identity. Log axes where the fitting bounds span 100x or more.

Figure C1

Parameter recovery. Each panel is one parameter of one model; each point is one synthetic dataset, generated at the value on the x axis and fit by the same model. Points on the dashed identity line are recovered exactly.

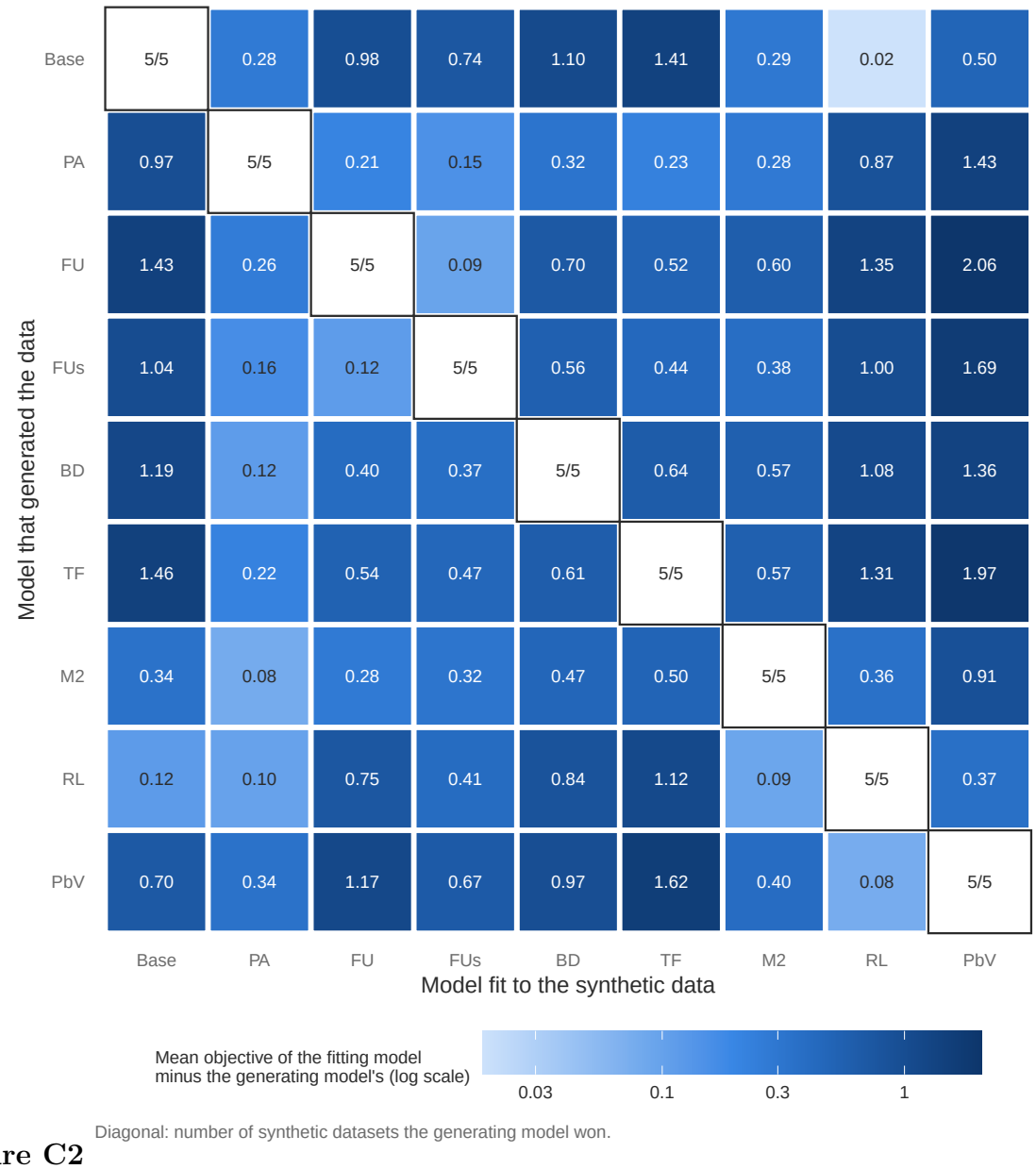


Figure C2

Model recovery. Rows: the model that generated the synthetic data (at its group-fit parameters); columns: the model fit to them. Off-diagonal cells give the mean amount by which the fitting model's objective exceeded the generating model's (darker = easier to tell apart); the diagonal gives the number of datasets the generating model won.

Table C1

Parameter recovery: correlation between generating and recovered values over 20 synthetic datasets per model, and the median absolute error, both on the scale shown (natural log for log-scaled parameters).

Model	Parameter	Scale	r	Median abs. error
PA	λ	log	.90	0.69
PA	β	log	.99	0.099
FU	χ	log	1.00	0.026
FU	λ	log	1.00	0.039
FU	α	linear	1.00	0.0011
FUs	χ	log	.99	0.17
FUs	λ	log	.78	0.19
FUs	α	linear	.99	0.0073
BD	α	log	.83	0.065
BD	δ	log	.80	0.12
BD	τ	log	.74	0.14
TF	x	linear	.96	0.014
TF	b	linear	.93	0.014
TF	α_0	linear	.99	0.006
M2	X	log	.98	0.064
RL	η	linear	.88	0.058
RL	β	log	.80	0.35
PbV	α	linear	.33	0.18
PbV	ϵ	linear	.82	0.15

variants' improvement comes from the choice rule alone.

These models were recovered in a separate run with the same procedure (Table C2; Figures C4–C5): 20 parameter draws per model, and model recovery among the 8 models Base, PA, FU, Base-S, R-W, R-W-S, R-W-D, and KF-F, with 5 synthetic datasets per generator. All are deterministic, so every fit used the deterministic budget.

The softmax models are identifiable. Every parameter of Base-S, R-W-S, R-W-D, and KF-F was recovered with $r \geq .95$, except KF-F's process noise τ^2 ($r = .80$),

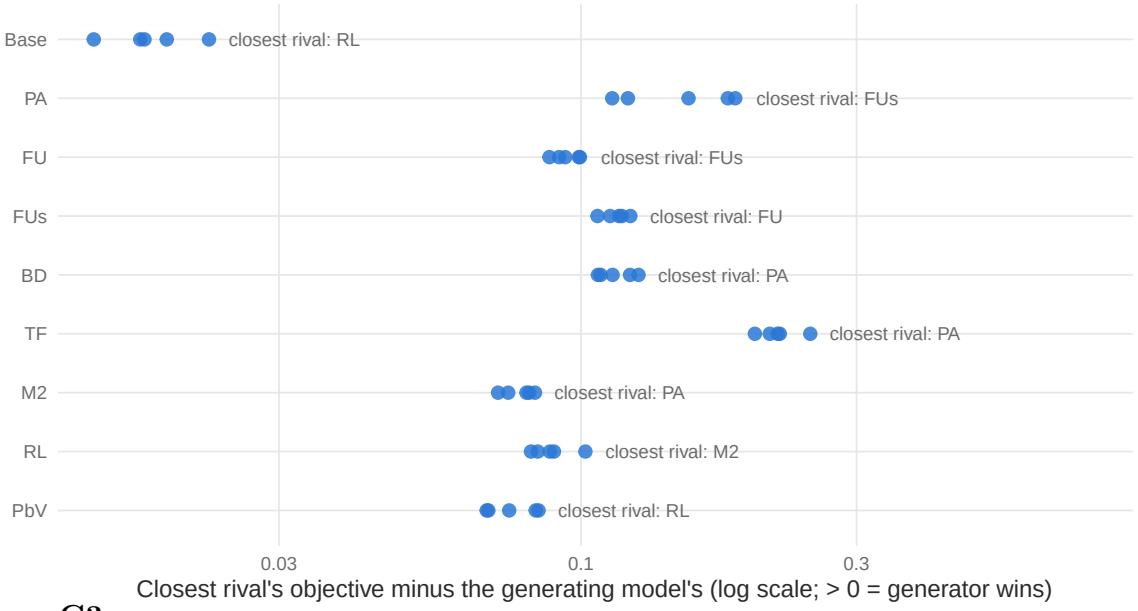


Figure C3

Margin of victory: for each synthetic dataset, how much worse the best-fitting rival model's objective was than the generating model's, with the rival that came closest most often.

whose two large misses came from draws with very small initial uncertainty σ_0^2, where diffusion has little effect on the predictions. In model recovery, each of them won all 5 of its own datasets, by margins of .01–.05 over its closest rival, usually another error-driven softmax model. These margins are smaller than those separating the main models (at least .07 above), so the 47 designs distinguish the error-driven variants from one another, but only just.

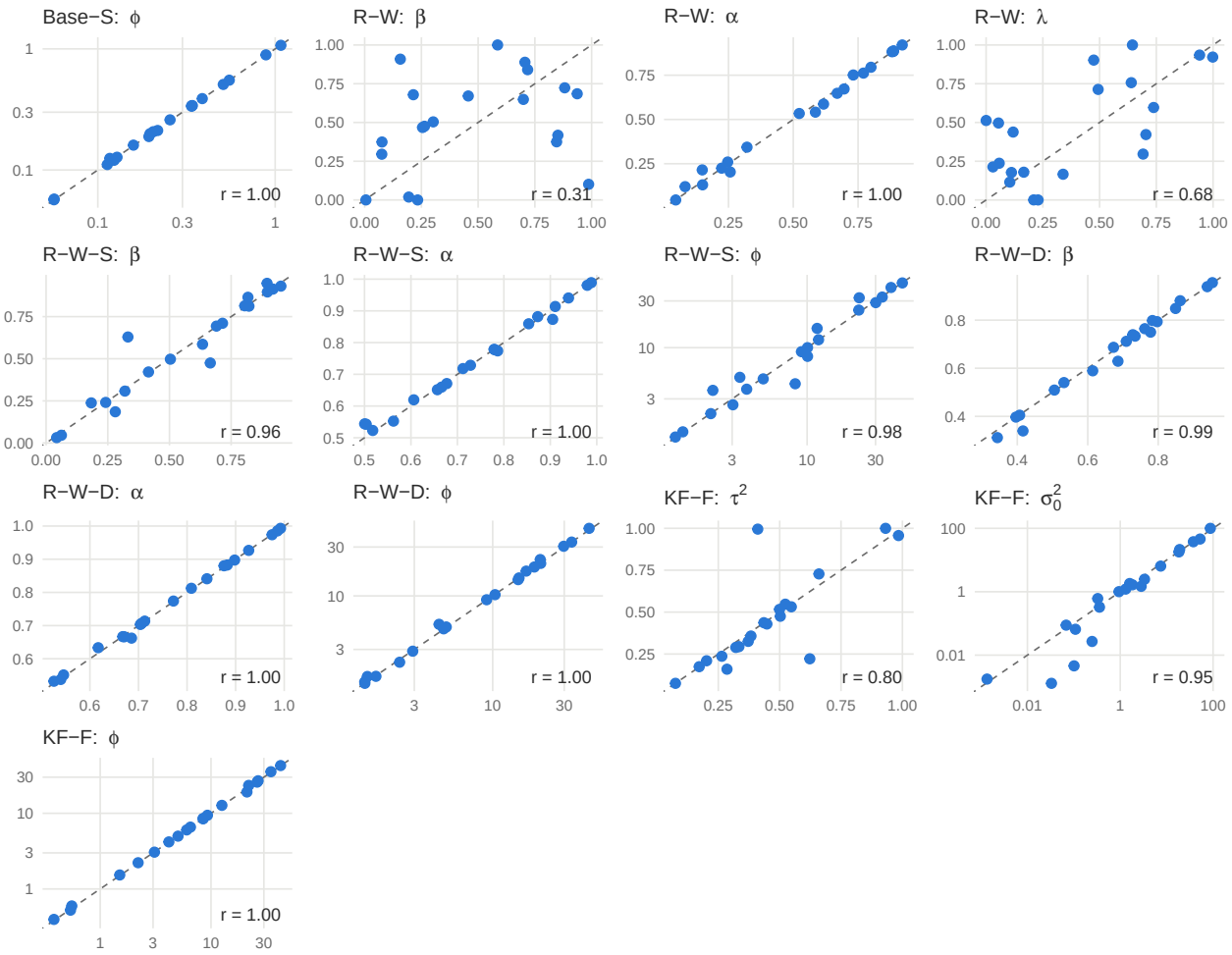
The corrected Rescorla-Wagner model with Luce choice is not. Its decay α is recovered ($r = 1.00$), but its learning rate β ($r = .31$) and asymptote λ ($r = .68$) are not, nor is their product, the effective learning rate ($r = .37$ on a log scale); 15% of fits put one of them on a fitting bound. In model recovery, R-W and the baseline cannot be told apart: on each other's data their objectives differed by at most .0009, and the baseline won only 1 of its own 5 datasets, the rest going to R-W's extra free parameters. Under Luce choice, then, the learning rate barely changes this model's predictions on these designs, and its fit should

be read as the baseline's. The error-driven mechanism is distinguishable only with a choice rule that is sensitive to the scale of the learned associations.

Table C2

Parameter recovery for the error-driven models, as in Table C1: 20 synthetic datasets per model.

Model	Parameter	Scale	r	Median abs. error
Base-S	ϕ	log	1.00	0.0073
R-W	β	linear	.31	0.22
R-W	α	linear	1.00	0.016
R-W	λ	linear	.68	0.2
R-W-S	β	linear	.96	0.011
R-W-S	α	linear	1.00	0.0059
R-W-S	ϕ	log	.98	0.036
R-W-D	β	linear	.99	0.0059
R-W-D	α	linear	1.00	0.001
R-W-D	ϕ	log	1.00	0.016
KF-F	τ^2	linear	.80	0.027
KF-F	σ_0^2	log	.95	0.14
KF-F	ϕ	log	1.00	0.01



x: generating (true) value; y: recovered value; dashed: identity. Log axes where the fitting bounds span 100x or more.

Figure C4

Parameter recovery for the error-driven models. As in Figure C1.

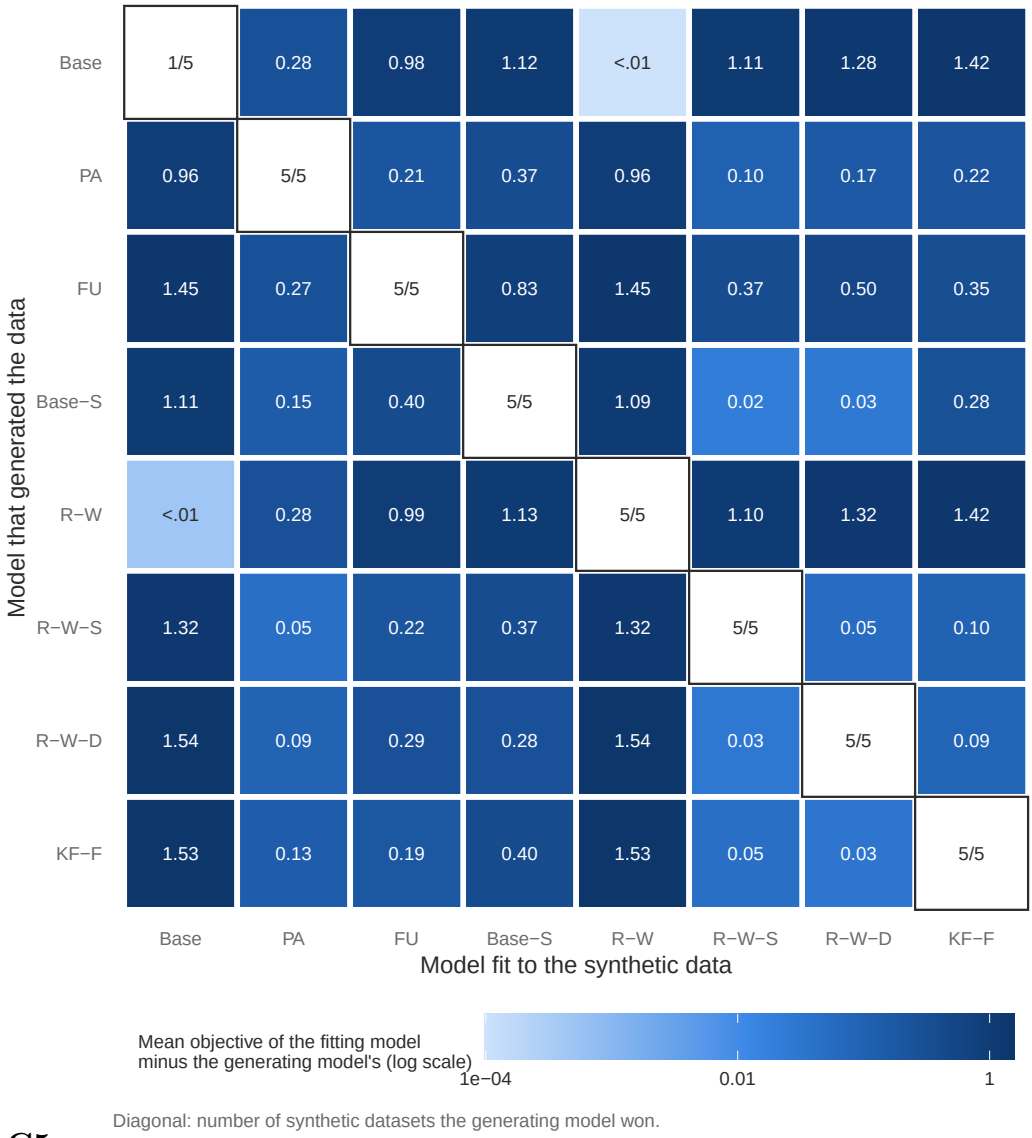


Figure C5

Model recovery for the error-driven models. As in Figure C2. Base and R-W are indistinguishable (gaps under .01 in both directions); every other generator won all its datasets.